

LU EXamples

1. Use LU Method to solve the following equation system.

$$\begin{array}{rrcr} x_1 & + & 5x_2 & - & 5x_3 & = & 12 \\ 2x_1 & + & 3x_2 & + & x_3 & = & 14 \\ x_1 & + & 4x_2 & - & 2x_3 & = & 12 \end{array}$$

$$\begin{bmatrix} 1 & 5 & -5 \\ 2 & 3 & 1 \\ 1 & 4 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 12 \\ 14 \\ 12 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 5 & -5 \\ 2 & 3 & 1 \\ 1 & 4 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

$$u_{11} = 1, u_{12} = 5, u_{13} = -5$$

$$l_{21}u_{11} = 2, \quad l_{21} = 2$$

$$l_{21}u_{12} + u_{22} = 3, \quad u_{22} = -7$$

$$l_{21}u_{13} + u_{23} = 1, \quad u_{23} = 11$$

$$l_{31}u_{11} = 1, \quad l_{31} = 1$$

$$l_{31}u_{12} + l_{32}u_{22} = 4, \quad l_{32} = 0.142$$

$$l_{31}u_{13} + l_{32}u_{23} + u_{33} = -2, \quad u_{33} = 0.42857$$

L Matrix:

1.00000	0.00000	0.00000
2.00000	1.00000	0.00000
1.00000	0.14286	1.00000

U Matrix:

1.00000	5.00000	-5.00000
0.00000	-7.00000	11.00000
0.00000	0.00000	1.42857

$$Ax=B \quad LUX=B \quad Ux=Y \quad LY=B$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 0.14286 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 12 \\ 14 \\ 12 \end{bmatrix},$$

$$y_1 = 12 \quad y_2 = 14 - 2 * 12 = -10, \quad y_3 = 12 - 12 - 0.14286 * (-10) = 1.4286$$

Y Solution:

12.00000	-10.00000	1.42857
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$$\begin{bmatrix} 1 & 5 & -5 \\ 0 & -7 & 11 \\ 0 & 0 & 1.42857 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 12 \\ -10 \\ 1.4286 \end{bmatrix}$$

$$x_3 = 1, x_2 = \frac{-10 - 11}{7} = 3 \quad x_1 = 12 - 5 * 3 + 5 = 2$$

X Solution:

2.00000	3.00000	1.00000
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2. Use LU Method to solve the following equation system.

$$2x_1 - x_2 + x_3 = -1,$$

$$3x_1 + 3x_2 + 9x_3 = 0,$$

$$3x_1 + 3x_2 + 5x_3 = 4.$$

L Matrix:

1.00000	0.00000	0.00000
1.50000	1.00000	0.00000
1.50000	1.00000	1.00000

U Matrix:

2.00000	-1.00000	1.00000
0.00000	4.50000	7.50000
0.00000	0.00000	-4.00000

Y Solution:

-1.00000	1.50000	4.00000
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X Solution:

1.00000	2.00000	-1.00000
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3. Use LU Method to solve the following equation system.

$$1.012x_1 - 2.132x_2 + 3.104x_3 = 1.984,$$

$$-2.132x_1 + 4.096x_2 - 7.013x_3 = -5.049,$$

$$3.104x_1 - 7.013x_2 + 0.014x_3 = -3.895.$$

L Matrix:

1.00000	0.00000	0.00000
-2.10672	1.00000	0.00000
3.06719	1.19776	1.00000

U Matrix:

1.01200	-2.13200	3.10400
0.00000	-0.39553	-0.47374
0.00000	0.00000	-8.93914

Y Solution:

1.98400	-0.86927	-8.93914
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X Solution:

1.00000	1.00000	1.00000
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4. Use LU Method to solve the following equation system.

$$\begin{aligned}2x_1 &= 3, \\x_1 + 1.5x_2 &= 4.5, \\-3x_2 + 0.5x_3 &= -6.6, \\2x_1 - 2x_2 + x_3 + x_4 &= 0.8.\end{aligned}$$

L Matrix:

1.00000	0.00000	0.00000	0.00000
0.50000	1.00000	0.00000	0.00000
0.00000	-2.00000	1.00000	0.00000
1.00000	-1.33333	2.00000	1.00000

U Matrix:

2.00000	0.00000	0.00000	0.00000
0.00000	1.50000	0.00000	0.00000
0.00000	0.00000	0.50000	0.00000
0.00000	0.00000	0.00000	1.00000

Y Solution:

3.00000	3.00000	-0.60000	3.00000
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X Solution:

1.50000	2.00000	-1.20000	3.00000
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